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Studierichting: Informatica
Oefeningenreeks 4A nr. 38.6

$$\begin{aligned} & \int \frac{5x-7}{x^2+3x+3} dx \\ &= \frac{5}{2} \int \frac{2x+3}{x^2+3x+3} dx - \frac{29}{2} \int \frac{dx}{x^2+3x+3} \end{aligned}$$

Deel 1:

$$\frac{5}{2} \int \frac{2x+3}{x^2+3x+3} dx$$

$$t = x^2 + 3x + 3$$

$$dt = (2x+3)dx$$

$$\begin{aligned} &= \frac{5}{2} \int \frac{dt}{t} \\ &= \frac{5}{2} \int \frac{dt}{t} \\ &= \frac{5}{2} \ln(t) + C \\ &= \frac{5}{2} \ln(x^2 + 3x + 3) + C \end{aligned}$$

Deel 2:

$$\left(x + \frac{3}{2}\right)^2 + \frac{3}{4} = x^2 + 3x + 3$$

$$\begin{aligned} & -\frac{29}{2} \int \frac{dx}{x^2+3x+3} \\ &= -\frac{29}{2} \int \frac{dx}{\left(x + \frac{3}{2}\right)^2 + \frac{3}{4}} \end{aligned}$$

$$u = x + \frac{3}{2}$$

$$du = dx$$

$$= -\frac{29}{2} \int \frac{du}{u^2 + \frac{3}{4}}$$

$$\begin{aligned}
&= -\frac{29}{\sqrt{3}} \operatorname{Bgtan}\left(\frac{2u}{\sqrt{3}}\right) + C \\
&= -\frac{29}{\sqrt{3}} \operatorname{Bgtan}\left(\frac{2x+3}{\sqrt{3}}\right) + C
\end{aligned}$$

Deel 1 + Deel 2:

$$= \frac{5}{2} \ln(x^2 + 3x + 3) - \frac{29}{\sqrt{3}} \operatorname{Bgtan}\left(\frac{2x+3}{\sqrt{3}}\right) + C$$

References